ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 2 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Practical work is not marked strictly on syntax. Give marks for correct logic, correct pins and connections, and neat, labelled work. Accept other layouts, ladder programs and program structures that work.

QUESTION 1

1.1 HX711 VCC and GND ✓, DT and SCK to digital pins ✓✓; load cell wires (E+, E−, A+, A−) to the HX711 ✓; LCD I2C module VCC, GND ✓, SDA to A4 and SCL to A5 ✓✓; ULN2003 IN1–IN4 to four digital pins ✓, driver powered with 5 V and GND ✓, stepper plugged into the driver ✓; push button to a pin and GND ✓; LEDs with 220 Ω resistors ✓; buzzer to a pin and GND ✓; neat, no short circuits ✓ (14)

1.2 Start ✓; set up pins, scale (tare), LCD "Place parcel" ✓✓; loop ✓; decision: WEIGH pressed? ✓; read mass and display ✓; decision: mass < 0,05 kg? (No parcel branch) ✓; decision: mass > 5 kg? ✓; Yes – "Too heavy", red LED, three beeps, reject count + 1 ✓✓; No – decision: mass ≤ 2 kg? band A or band B, count + 1 ✓; green LED on, open gate, wait 3 s, close gate, LED off ✓✓; print counters and loop back ✓ (14)

1.3 Circuit ✓ and flowchart ✓ saved and printed / presented (2)

[30]

QUESTION 2

2.1 Libraries (HX711, LCD I2C, Stepper) and objects ✓✓; pin modes incl. INPUT_PULLUP ✓✓; Serial.begin ✓; scale begin, calibration factor and tare ✓✓; LCD started and "Place parcel" ✓; button read and LOW tested ✓; mass read as an average of 10 readings ✓; mass shown on line 1 ✓; "No parcel" check ✓✓; band A / band B decision ✓✓ shown on line 2 ✓; green LED, gate opened 90° (512 steps) ✓, 3 s wait ✓, gate closed (−512 steps) ✓; too heavy: message, red LED ✓, three beeps in a loop ✓✓; three counters increased ✓✓✓ and printed ✓; wait for the button to be released ✓; comments ✓ (28)

```
#include <HX711.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Stepper.h>

HX711 scale;
LiquidCrystal_I2C lcd(0x27, 16, 2);
Stepper gate(2048, 8, 10, 9, 11);        // 2048 steps per turn, ULN2003 on pins 8-11
const int button = 2, buzzer = 6, redLed = 7, greenLed = 12;
int bandA = 0, bandB = 0, rejected = 0;

void setup() {
  pinMode(button, INPUT_PULLUP);
  pinMode(buzzer, OUTPUT); pinMode(redLed, OUTPUT); pinMode(greenLed, OUTPUT);
  Serial.begin(9600);
  scale.begin(3, 4);                     // DT, SCK
  scale.set_scale(21500);                // calibration factor found with a known mass
  scale.tare();                          // empty scale reads 0
  gate.setSpeed(10);                     // r/min
```

```
  lcd.init(); lcd.backlight();
  lcd.print("Place parcel");
}

void loop() {
  if (digitalRead(button) == LOW) {      // WEIGH pressed
    float mass = scale.get_units(10);    // average of 10 readings, in kg
    lcd.clear();
    lcd.print("Mass: "); lcd.print(mass, 2); lcd.print(" kg");
    lcd.setCursor(0, 1);
    if (mass < 0.05) {
      lcd.print("No parcel");
    } else if (mass > 5) {               // too heavy: reject
      lcd.print("Too heavy");
      digitalWrite(redLed, HIGH);
      for (int i = 0; i < 3; i++) {      // three beeps
        digitalWrite(buzzer, HIGH); delay(300);
        digitalWrite(buzzer, LOW);  delay(300);
      }
      digitalWrite(redLed, LOW);
      rejected++;
    } else {
      if (mass <= 2) { lcd.print("Band A"); bandA++; }
      else           { lcd.print("Band B"); bandB++; }
      digitalWrite(greenLed, HIGH);
      gate.step(512);                    // open: a quarter turn = 90 degrees
      delay(3000);
      gate.step(-512);                   // close
      digitalWrite(greenLed, LOW);
    }
    Serial.print("A: "); Serial.print(bandA);
    Serial.print("  B: "); Serial.print(bandB);
    Serial.print("  Rejected: "); Serial.println(rejected);
    while (digitalRead(button) == LOW) {} // wait for the button to be released
  }
}
```

2.2 Saved ✓ and printed / presented ✓ (2)

[30]

QUESTION 3

3.1 Weighing station ✓; conveyor from the station to the robot ✓; robot ✓ with its work envelope drawn (circle / arc) ✓; stacking area (pallet) inside the envelope ✓; safety fence around the robot and stacking area ✓ with an interlocked access gate ✓; light curtain where the conveyor enters the fence ✓; emergency stops at the operator's position and at the gate ✓✓; operator's position outside the fence ✓; labels and neat drawing ✓ (12)

3.2 START I0.0 (NO) ✓ in parallel with a Q0.0 holding contact ✓; in series with STOP I0.1 ✓, E-stop I0.2 ✓ and light curtain I0.3 ✓ (all as normally closed devices, programmed so the rung breaks when they open) ✓✓; driving the conveyor coil Q0.0 ✓; latch drops out when any stop condition occurs, so START is needed again ✓; rung with Q0.0 contact driving the green lamp Q0.1 ✓✓; rung with a NC Q0.0 contact driving the red lamp Q0.2 ✓✓; correct ladder symbols, addresses and labels ✓✓, e.g. (15)

```
|─┬─[ I0.0 ]─┬─[ I0.1 ]──[ I0.2 ]──[ I0.3 ]───( Q0.0 )──|   conveyor (latched)
| └─[ Q0.0 ]─┘                                      |
|──[ Q0.0 ]──────────────────────────────────( Q0.1 )──|   green: running
|──[/ Q0.0 ]──────────────────────────────────( Q0.2 )──|   red: stopped
(STOP, E-stop and light curtain are NC devices, so their inputs are ON while healthy.)
```

3.3 Three sensible hazards ✓✓✓; a matching injury or damage, a sensible risk level and a suitable control measure for each hazard ✓✓✓✓✓✓ (any seven of the nine remaining cells correct), e.g. robot arm hitting a person – crushing / serious injury – high – fence, interlocked gate, light curtain, E-stops; trapping fingers in the conveyor – cuts, crushed fingers – medium – guards over rollers and nip points, E-stop pull cord; heavy parcels falling from the stack – foot injury, damaged goods – medium – stack height limit, safety boots, stable pallet; tripping over cables – falls – low – cable trays and covers; electric shock – burns, death – high – enclosed panels, lockout, qualified electricians. (10)

3.4 Layout ✓, ladder program ✓ and risk assessment ✓ saved and printed / presented (3)

[40]

TOTAL: 100

END OF MARKING GUIDELINE